



Validating Causality

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PO

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PO

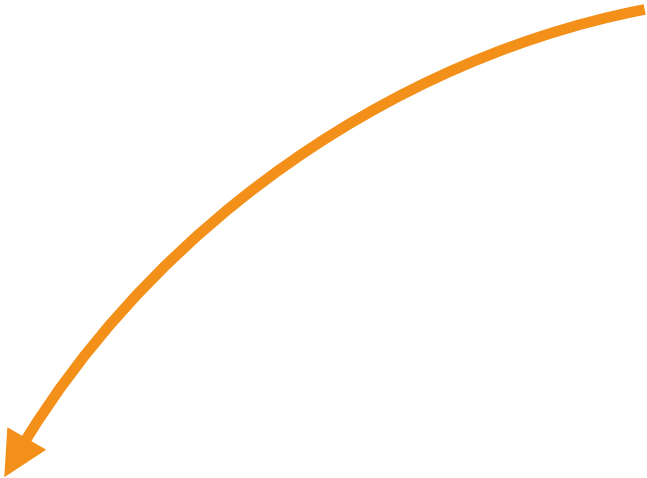
P

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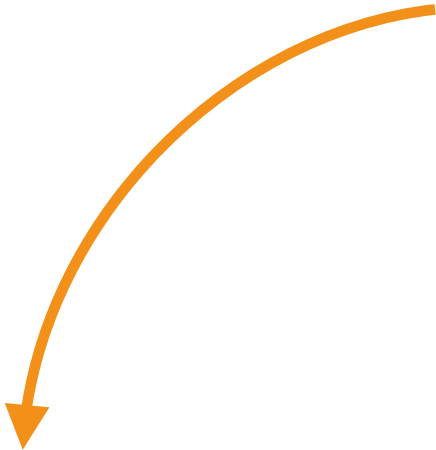


po

po



cosrc



collog



S



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R  +

R



-



S



-



C



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R  +

R  -



C



—

```
update(src, +100)
```

```
update(src, -100)
```



```
def update(account: cown[Account], amount: int) {  
  when(account) { A  
    if(account.balance + amount > 0) {  
      account.balance += amount  
      val id = account.id()  
      when(log) { B log.record(id, amount) }  
    } } }  
}
```

We are given the program events

We are given the program and constraints

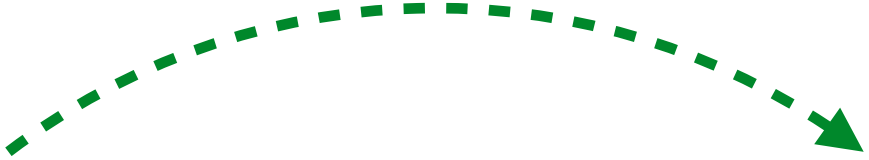
We derive the run and happens before orders

Wettest for acidity

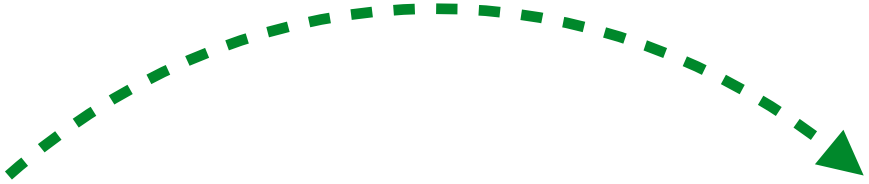




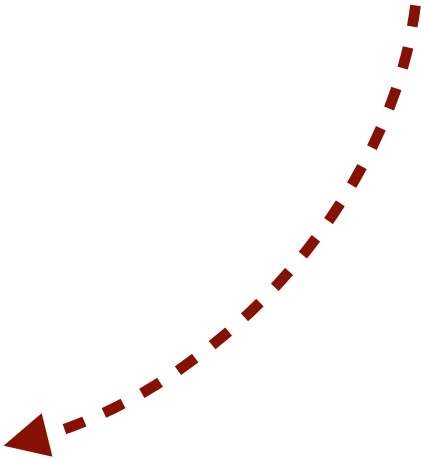


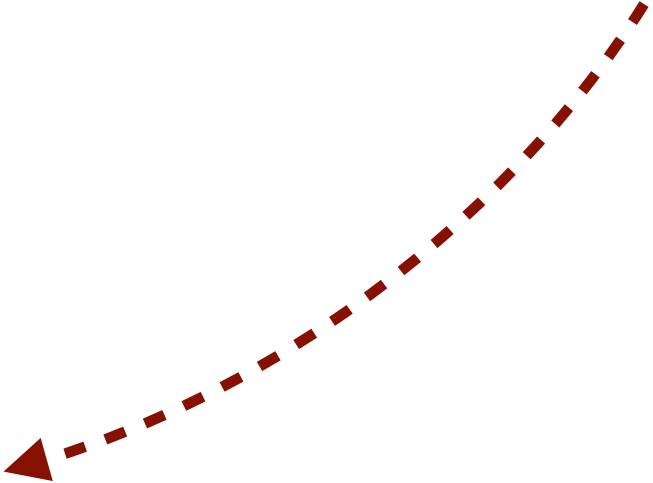












hb

h

b

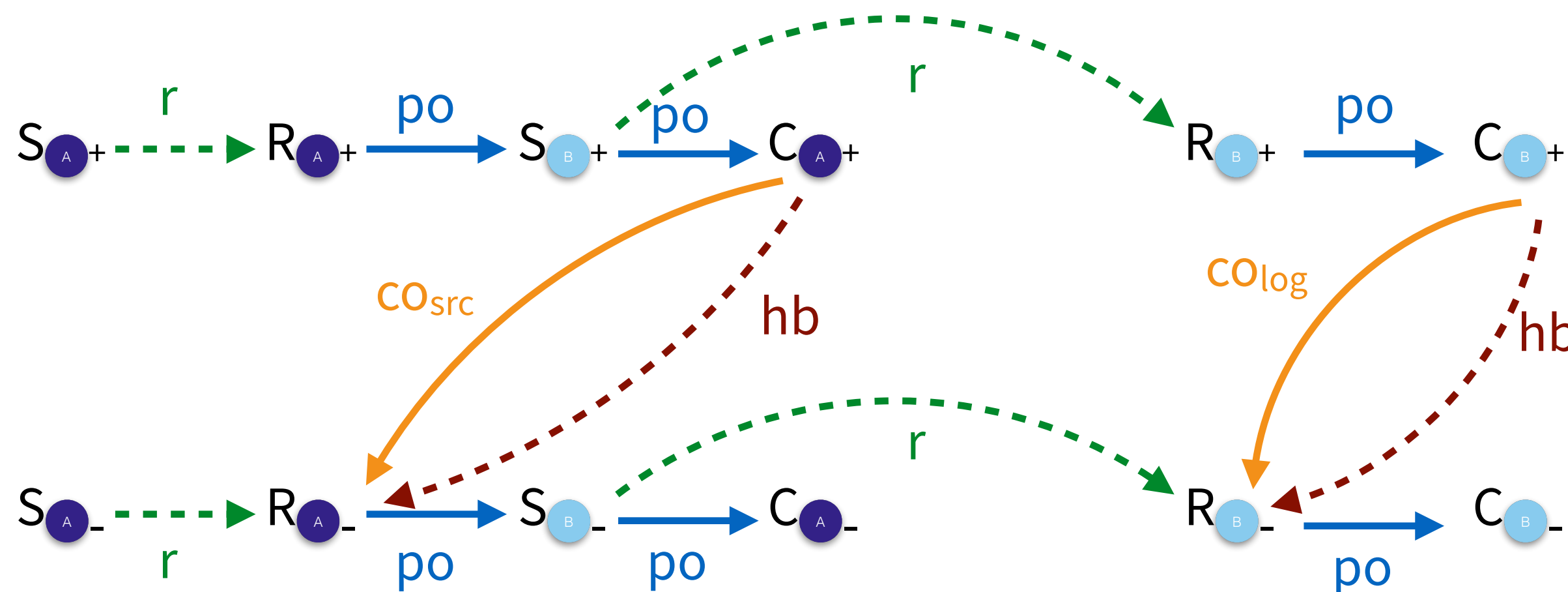


Validating Causality

```
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    if(account.balance + amount > 0) {  
      account.balance += amount  
      val id = account.id()  
      when(log) { B log.record(id, amount) }  
    }  
  }  
}
```

update(src, +100)

update(src, -100)



We are given the program events

We are given the program and cown orders

We derive the run and happens before orders

We test for acyclicity ✓

Validating Causality

```
def update(account: cown[Account], amount: int) {  
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update(src, +100)

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